Tutorial 1 (Solutions)

EC3301 - Martinmas Semester

Statistical inference

Consider a random sample (iid: independently and identitically distributed) drawn from the distribution

$$y_i \sim \mathcal{N}(2, 25)$$

1. What is the $E[y_i]$ and $Var(y_i)$?

$$\begin{aligned} E[y_i] &= 2 \\ Var(y_i) &= 25 \end{aligned}$$

2. Solve for the expected value and variance of the sample mean: $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$.

$$\begin{aligned} E[\bar{y}] =& E\left[\frac{1}{n}\sum_{i=1}^{n} y_i\right] \\ =& \frac{1}{n}\sum_{i=1}^{n} E[y_i] \\ =& \frac{1}{n}\sum_{i=1}^{n} 2 \\ =& \frac{2n}{n} \\ =& 2 \end{aligned}$$

$$
\begin{aligned}
Var(\bar{y}) =& E[(\bar{y} - \underbrace{E[\bar{y}]}_{=2})^2] \\
=& E\left[\left(\frac{1}{n}\sum_{i=1}^{n}(y_i - 2)\right)^2\right] \\
=& E\left[\frac{1}{n^2}\sum_{i=1}^{n}(y_i - 2)^2 + 2 \times \frac{1}{n^2}\sum_{i=1}^{n-1}\sum_{j=i+1}^{n}(y_i - 2)(y_j - 2)\right] \\
=& \frac{1}{n^2}\sum_{i=1}^{n}\underbrace{E[(y_i - 2)^2]}_{Var(y_i)} + 2 \times \sum_{i=1}^{n-1}\sum_{j=i+1}^{n}\underbrace{E[(y_i - 2)(y_j - 2)]}_{Cov(y_i, y_j)=0} \\
=& \frac{nVar(y_i)}{n^2} \\
=& \frac{25}{n}
\end{aligned}
$$

3. What is the distribution of $\bar{y}$?

$$\bar{y} \sim \mathcal{N}(2, 25/n)$$

4. Compute the probability that $Pr(\bar{y} < 3)$ in a sample of $n = 100$.

Since $\bar{y} \sim \mathcal{N}(2, 1/4)$,

$$\frac{\bar{y} - 2}{0.5} \sim \mathcal{N}(0, 1)$$

$$
\begin{aligned}
Pr(\bar{y} < 3) =& Pr(\bar{y} - 2 < 3 - 2) \\
=& Pr\left(\frac{\bar{y} - 2}{0.5} < \frac{1}{0.5}\right) \\
=& Pr(z < 2)
\end{aligned}
$$

```
dis normal(2)
```

```
Running C:\Users\ndl1\OneDrive - University of St Andrews\Documents\EC3301\webs
> ite\st-andrews-ec3301\profile.do ...
```

```
<IPython.core.display.HTML object>
```

```
.97724987
```

Suppose, you draw a random sample that has the following properties. For the next questions we will assume no knowledge of the true mean of y; only that it the sample was drawn from a normal distribution $y_i \sim \mathcal{N}(\mu, \sigma^2)$

```
sum y
mean y
```

```
    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
           y |        100    2.218156    2.082121   -2.61975   6.654652

Mean estimation                           Number of obs = 100

--------------------------------------------------------------
             |       Mean   Std. err.     [95% conf. interval]
-------------+------------------------------------------------
           y |   2.218156   .2082121      1.805018    2.631294
--------------------------------------------------------------
```

5. What is the relationship between the standard deviation of the sample and the standard error of the mean reported above?

The standard deviation (s) is the square-root of the sample variance. The sample variance is an estimate of the unobserved distribution variance ($Var(y_i) = 4$). In our above example $s = 2.082121$ which is $\approx 2 = \sqrt{4}$. The standard error (se) is the square root of the $Var(\bar{y})$. As we proved above, $Var(\bar{y}) = Var(y_i)/n$. That means that $se(\bar{y}) = sd/\sqrt{n}$. As $n = 100$, we find that $se = .2082121$; exactly equal to the $s/10$.

6. Earlier we showed that, $\frac{\bar{y}-2}{se(\bar{y})} \sim \mathcal{N}(0,1)$ where $se(\bar{y}) = \sigma/\sqrt{n}$. Since we do not know σ, what happens to the distribution if we replace σ with s, the standard deviation of the sample.

$$\frac{\bar{y}-2}{s/\sqrt{n}} \sim T_{n-1}$$

7. Using the properties of the sample, test the null hypothesis $H_0 : \mu = 1.85$ (where μ is the unobserved $E[y_i]$). Use a significance level of $\alpha = 0.05$.

$$\frac{\bar{y}-1.85}{s/\sqrt{n}} \underset{H_0}{\sim} T_{n-1}$$

Compute the test statistic:

```
dis (_b[y]-1.85)/_se[y]
```

```
1.7681755
```

> **Note**
>
> The `mean` command stores estimates in a similar way to `regress`. They are both e-class commands.
>
> - The `_b[]` function can be used to call the stored coefficient estimate for a particular variable. The full set of coefficients is stored as in the vector `e(b)`.
> - The `_se[]` function can be used to call the SE of a particular coefficient. These computed using the covariance matrix stored as `e(V)`.
>
> To see the full list of stored estimates, type `ereturn list`. Note, this does not work for `summarize`, which is a r-class command. After summarize you can run `return list` to see the format of its stored estimates.

Rejection rule:

- Since this is a two-sided test, we will reject if $|\text{test-stat}| < \text{t}_{99,0.975}$.

The test static remains the same.

Critical values of t-distribution (which is symmetric):

```
dis invt(99,0.975)
```

```
1.984217
```

Since $|\text{test-stat}| < \text{t}_{99,0.975}$, we fail to reject.

8. Using the properties of the sample, test the null hypothesis $H_0 : \mu \leq 1.85$.

Rejection rule:

- Since this is a one-sided test, we will reject if $\text{test-stat} > \text{t}_{99,0.95}$.

Critical values of t-distribution (which is symmetric):

```
dis invt(99,0.95)
```

```
1.6603912
```

We reject H_0 and conclude that $\mu > 1.85$.

9. Why is the conclusion different for the two hypotheses?

The only difference is the critical value. The one-side test has a lower critical value than the two-sided test.

10. Which test has more statistical power? Compute the power of the test to reject H_0 when the true value of $\mu = 2$.

In the two-sided test:

$$\begin{aligned}
Pr(\text{Reject } H_0|\mu = 2) =& Pr\left(\left|\frac{\bar{y} - 1.85}{s/\sqrt{n}}\right| > \mathrm{t}_{99,0.975}\middle|\mu = 2\right) \\
=& Pr\left(\frac{\bar{y} - 1.85}{s/\sqrt{n}} < \mathrm{t}_{99,0.025}\middle|\mu = 2\right) + Pr\left(\frac{\bar{y} - 1.85}{s/\sqrt{n}} > \mathrm{t}_{99,0.975}\middle|\mu = 2\right) \\
=& Pr\left(\frac{\bar{y} - 2}{s/\sqrt{n}} < \mathrm{t}_{99,0.025} - \frac{0.15}{s/\sqrt{n}}\middle|\mu = 2\right) + Pr\left(\frac{\bar{y} - 2}{s/\sqrt{n}} > \mathrm{t}_{99,0.975} - \frac{0.15}{s/\sqrt{n}}\middle|\mu = 2\right) \\
=& T_{n-1}(\mathrm{t}_{99,0.025} - \frac{0.15}{s/\sqrt{n}}) + 1 - T_{n-1}(\mathrm{t}_{99,0.975} - \frac{0.15}{s/\sqrt{n}})
\end{aligned}$$

```
dis t(99,invt(99,0.025)-0.15/_se[y]) + 1 - t(99,invt(99,0.975)-0.15/_se[y])
```

```
.10866073
```

In the one-sided test:

$$\begin{aligned}
Pr(\text{Reject } H_0|\mu = 2) =& Pr\left(\frac{\bar{y} - 1.85}{s/\sqrt{n}} > \mathrm{t}_{99,0.95}\middle|\mu = 2\right) \\
=& Pr\left(\frac{\bar{y} - 2}{s/\sqrt{n}} > \mathrm{t}_{99,0.95} - \frac{0.15}{s/\sqrt{n}}\middle|\mu = 2\right) \\
=& 1 - T_{n-1}(\mathrm{t}_{99,0.95} - \frac{0.15}{s/\sqrt{n}})
\end{aligned}$$

```
dis 1 - t(99,invt(99,0.95)-0.15/_se[y])
```

```
.17475989
```

This shows that the one-sided test is a more powerful test.

OLS estimator

In each case, try to prove the following properties of the OLS estimators $(\hat{\beta}_0, \hat{\beta}_1)$ for a simple linear regression model

$$y_i = \beta_0 + \beta_1 x_i + u_i$$

a. $\sum_{i=1}^{n} \hat{u}_i = 0$, where $\hat{u}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$

b. $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$, where $\bar{y} = \frac{1}{n} \sum_{i=1}^{n} y_i$ and $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$.

c. $\sum_{i=1}^{n} x_i \hat{u}_i = 0$

d. $\widehat{Cov}(x_i, \hat{u}_i) = 0$ where $\widehat{Cov}()$ is the sample covariance

e. $\widehat{Cov(y_i, \hat{y}_i)} = \widehat{Var}(x_i)$ where $\widehat{Var}()$ is the sample variance

Question 2

Question 3